

Parameters: vocab_size = 32000
input_embedding_dim = 4096
attention_head = 32
attention_layer = 32
attention_dim = 128
Feed-Forward-Dim = 11008

16QA: 4 key/value groups with 8 query each
(Group Query Attention)

① input_embedding = 32000×4096

②

$\Sigma \text{ RMS Norm} = 4096 \times 2 = 8192$ // 兩層

$W_Q = 4096 \times 128 \times 32 = 16,777,216$

$W_K = W_V = 4096 \times 128 \times 4 = 2,097,152$

$W_O = 4096 \times 4096 = 16,777,216$

+ FeedForward = $3 \times 4096 \times 11008 = 135,266,304$

173,023,232

③ 4096

④ 4096×32000

total = ① + ② $\times 32$ + ③ + ④

= 5,798,891,520

✗

